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Bacteriology MCQs With Answers: Practice Questions by Topic

Bacteriology multiple choice questions with answers and explanations, organized by topic: lab diagnosis, resistance mechanisms, skin and eye infections, zoonoses, STIs, and pathogen characteristics.



Acharya Tankeshwar

MSc (Medical Microbiology)

#Bacteriology MCQs

Practice bacteriology multiple choice questions with answers and explanations. Start with the ten questions below, then continue with more MCQs organized by topic.





MCQs Bacteriology (1-10)



- The identification of bacteria by serologic tests is based on the presence of specific antigens. Which of the following bacterial components is least likely to contain useful antigens?
 - Capsule
 - Cell wall
 - Flagella
 - Ribosomes
- Each of the following statements concerning the Gram stain is correct except:
 - Escherichia coli* stains pink because it has a thin peptidoglycan layer
 - Streptococcus pyogenes* stains blue because it has a thick peptidoglycan layer
 - Mycoplasma pneumoniae* is not visible in the Gram's stain because it does not have a cell wall
 - Mycobacterium tuberculosis* stain blue because it has a thick lipid layer
- An outbreak of sepsis caused by *Staphylococcus aureus* has occurred in the newborn nursery. You are called upon to investigate. According to your knowledge of the normal flora, what is the most likely source of the organism?
 - Nose
 - Colon
 - Hand
 - Throat
- Each of the following organisms is an important cause of urinary tract infections except:
 - Klebsiella pneumoniae*
 - Escherichia coli*

- c. *Bacteroides fragilis*
- d. *Proteus mirabilis*

5. A 30-year woman has non-bloody diarrhea for the past 14 hours. Which one of the following organisms is least likely to cause this illness?

- a. *Streptococcus pyogenes*
- b. *Clostridioides difficile*
- c. *Shigella dysenteriae*
- d. *Salmonella enteritidis*

6. Each of the following agents is a recognized cause of diarrhea except

- a. *Clostridium perfringens*
- b. *Vibrio cholerae*
- c. *Enterococcus faecalis*
- d. *Escherichia coli*

7. Each of the following statements about the classification of streptococci is correct except

- a. Pneumococci (*Streptococcus pneumoniae*) are alpha-hemolytic and can be serotyped on the basis of their polysaccharide capsule
- b. Enterococci are group D streptococci and can be classified by their ability to grow 6.5% sodium chloride
- c. Viridans streptococci are identified by Lancefield grouping, which is based on the C carbohydrate in the cell wall
- d. Although pneumococci and the viridans streptococci are alpha-hemolytic, they can be differentiated by the bile solubility test and their susceptibility to optochin

8. Which of the following bacteria has the lowest 50% infective dose (ID₅₀)?

- a. *Campylobacter jejuni*
- b. *Salmonella typhi*
- c. *Vibrio cholerae*
- d. *Shigella sonnei*

9. Which of the following disease is best diagnosed by serologic means?

- a. Pulmonary tuberculosis
- b. Gonorrhea
- c. Actinomycosis
- d. Q Fever

10. The coagulase test is used to differentiate

- a. *Staphylococcus epidermidis* from *Neisseria meningitidis*
- b. *Staphylococcus aureus* from *Staphylococcus epidermidis*
- c. *Streptococcus pyogenes* from *Staphylococcus aureus*
- d. *Streptococcus pyogenes* from *Enterococcus faecalis*

Browse bacteriology MCQs by topic:

- [Laboratory Diagnosis of Bacterial Infections](#)
- [Virulence Factors, Resistance and Toxin Mechanisms](#)
- [Key Characteristics of Bacterial Pathogens](#)
- [Bacterial Skin and Eye Infections](#)
- [Zoonoses, STIs and Spirochetes](#)

Answer Key

1. d (Ribosomes)
2. d (*Mycobacterium tuberculosis* stains blue because it has a thick lipid layer)
3. a (Nose)
4. c (*Bacteroides fragilis*)
5. a (*Streptococcus pyogenes*)
6. c (*Enterococcus faecalis*)
7. c (Viridans streptococci are identified by Lancefield grouping)
8. d (*Shigella sonnei*)
9. d (Q fever)
10. b (*Staphylococcus aureus* from *Staphylococcus epidermidis*)

Why these are the answers

1. **Ribosomes.** Serologic identification relies on surface structures the immune system and antisera can reach: the capsule (K antigen), the cell wall O antigen, and the flagella (H antigen) are all classic serotyping targets. Ribosomes sit inside the cell and are highly conserved across species, so they carry no useful species-level antigenic signal. The pattern to remember: serology reads the outside of the cell, not the inside.
2. **The false statement is that *Mycobacterium tuberculosis* stains blue because of a thick lipid layer.** In fact the lipid (mycolic acid) layer is exactly why *M. tuberculosis* resists the Gram stain and does not stain reliably at all; it needs the [acid-fast \(Ziehl-Neelsen\) stain](#) instead. Options a, b, and c are all true: *E. coli* is gram-negative (thin peptidoglycan, stains pink), *S. pyogenes* is gram-positive (thick peptidoglycan, stains purple), and *Mycoplasma* has no cell wall so it cannot be seen on Gram stain at all. The trap is that "thick lipid layer" sounds like a real explanation; it is the reason for non-staining, not for blue staining.
3. **Nose.** This question asks for the reservoir in the normal flora, and the anterior nares are the primary carriage site for *S. aureus*: roughly 30% of healthy people carry it there. The clinically important subtlety, and why "Hand" is such a tempting distractor: in a nursery outbreak the

organism usually reaches the newborns on the hands of staff, but those hands are contaminated from the carrier's own nose. The nose is the source (reservoir); the hand is the vehicle (transmission route). The question wording, "according to your knowledge of the normal flora," points at the reservoir.

4. ***Bacteroides fragilis***. The question asks which is NOT an important cause of UTI. *E. coli*, *Klebsiella pneumoniae*, and *Proteus mirabilis* are all leading uropathogens; *Proteus* especially, because its urease raises urine pH and forms struvite stones. *Bacteroides fragilis* is a strict anaerobe of the gut and rarely causes UTI, because urine and the urinary tract are not an anaerobic environment. Remember: UTIs are overwhelmingly aerobic and facultative gram-negative rods; a strict anaerobe in the list is almost always the odd one out.
5. ***Streptococcus pyogenes***. The question asks which is LEAST likely to cause non-bloody diarrhea. *Clostridioides difficile*, *Shigella dysenteriae*, and *Salmonella* Enteritidis are all enteric pathogens that cause diarrhea. *S. pyogenes* is a throat and skin pathogen (pharyngitis, cellulitis, impetigo); it is not an enteric pathogen and does not cause gastroenteritis. The clue is anatomical: three of the four live in or attack the gut, one does not.
6. ***Enterococcus faecalis***. The question asks which is NOT a recognized cause of diarrhea. *Clostridium perfringens* (toxin-mediated food poisoning), *Vibrio cholerae* (watery cholera), and diarrheagenic *E. coli* are all established causes. *Enterococcus faecalis* is a normal gut commensal and a cause of UTI, endocarditis, and wound infection, but it is not a recognized diarrheal pathogen.
7. **The false statement is that viridans streptococci are identified by Lancefield grouping.** Lancefield grouping is based on the C carbohydrate in the cell wall and applies to beta-hemolytic streptococci (groups A, B, and so on). Viridans streptococci are alpha-hemolytic and lack a groupable C carbohydrate, so they are not Lancefield-typed; they are identified physiologically and biochemically instead. The other three statements are correct, including the key practical pair in option d: pneumococci and viridans strep are both alpha-hemolytic but are separated by the bile solubility test and optochin susceptibility (pneumococci are bile-soluble and optochin-sensitive; viridans are neither).
8. ***Shigella sonnei***. This asks for the lowest 50% infective dose. *Shigella* is famous for needing only a tiny inoculum, on the order of 10 to 100 organisms, because it survives stomach acid well and is highly efficient at invading the colonic mucosa. *Salmonella* Typhi, *Campylobacter jejuni*, and *Vibrio cholerae* all require far larger doses (typically thousands to millions) to establish infection. This low infective dose is exactly why *Shigella* spreads readily by person-to-person contact, unlike the food-and-water-dose-dependent spread of the others.
9. **Q fever.** *Coxiella burnetii*, the cause of Q fever, is diagnosed serologically because it is hazardous to culture (a biosafety risk) and does not grow on routine media. Pulmonary tuberculosis is diagnosed by smear, culture, and molecular tests; gonorrhea by culture or NAAT; actinomycosis by identifying sulfur granules and anaerobic culture. The rule of thumb: serology is the go-to when the organism is dangerous or impractical to culture, which fits *Coxiella* precisely.
10. **The coagulase test separates *Staphylococcus aureus* from *Staphylococcus epidermidis*.** Both are gram-positive cocci in clusters and both are catalase-positive, so they need a further test to tell apart: *S. aureus* is coagulase-positive (it clots plasma), while *S. epidermidis* and the other

skin staphylococci are coagulase-negative. The other pairings are wrong because those organisms are already separated by simpler tests: staphylococci versus streptococci by catalase, and *Neisseria* is a gram-negative diplococcus, distinguishable on Gram stain alone.